

Hamza Bouktitiya

Artificial Intelligence Engineer

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📍 Kenitra, Morocco 🔗 LinkedIn 🐙 GitHub

Education

- **PhD Candidate in Artificial Intelligence** *2024 – Present*
 - Ibn Tofail University, Kenitra, Morocco
 - Thesis: Textual description and automatic classification of video sequences by deep learning
 - Supervised by: Professor Housni Khalid
- **Master in Artificial Intelligence and Virtual Reality** *2022 – 2024*
 - Ibn Tofail University, Kenitra, Morocco
- **Bachelor in Mathematical and Computer Science** *2018 – 2022*
 - Ibn Tofail University, Kenitra, Morocco
- **Baccalaureate in Mathematical Sciences** *2017 – 2018*
 - Sidi Aissa High School, Souk El Arbaa Du Gharbe

Professional Experience

- **Junior AI Engineer** *April 2025 – October 2025*
 - **OBYSTECH SOLUTIONS, Rabat**
 - Designed a complete AI platform with Next.js frontend and FastAPI/LangChain backend
 - RAG pipeline: document ingestion, HF embeddings, hybrid/vector search with re-ranking (MMR)
 - LangGraph workflows with PostgreSQL-managed memory and MCP integration
 - Multi-service Docker Compose deployment (PostgreSQL + ChromaDB)
 - Technologies: Next.js, React, Tailwind, FastAPI, LangChain, LangGraph, MCP, Ollama, PostgreSQL, ChromaDB, Docker, JWT/OAuth2
- **AI Intern** *February 2024 – August 2024*
 - **Verve Technologie, Casablanca**
 - Built a forecasting system using LSTM to predict energy consumption
 - Set up image generation endpoints with Stable Diffusion 1.5
 - Developed a RAG system to generate text from Wikipedia and PDF
 - Added JWT authentication and user management
 - Technologies: LSTM, LangChain, FastAPI, Stable Diffusion, JWT
- **Web Development Intern** *March 2022 – June 2022*
 - **CS Department, Ibn Tofail University, Kenitra**
 - Created a web application for gym management (clients and administrators)
 - Technologies: PHP, JavaScript, Bootstrap, CSS, MySQL

Technical Skills

- **AI & Machine Learning:** PyTorch, PyTorch3D, TensorFlow, LangChain, LangGraph, CrewAI
- **Back-end:** Django, FastAPI, SQL, MySQL, PostgreSQL, PL/SQL
- **Front-end & Design:** HTML, CSS, Bootstrap, Next.js, THREE.JS, LangChain.js
- **Data Analysis:** Python, Power BI, Pandas, NumPy

Personal Projects

- **Chatbot_odoo** (2024) – AI assistant integrated with Odoo ERP for business operations management via NLP
- **Educational Game - DyslexAI** (February 2024) – Game to help children improve pronunciation and word learning
- **3D Object Generation** (February 2024) – Creating 3D objects from 2D images using NeRF
- **Health Chatbot - ThinkAI** (August 2024) – Chatbot using Python, Ollama and Streamlit
- **Book Recommendation System** (2024) – Collaborative filtering and machine learning algorithms
- **Darija Q&A System** (December 2023) – NLP system for question answering in Moroccan darija and English
- **License Plate Recognition** (June 2023) – Custom image recognition model with HOG, KNN and SVM (98.9% accuracy)
- **Gym-Explore** (December 2023) – Streamlit interface for Deep Q-Learning algorithms
- **Gym Management System** (March – June 2022) – PHP web application for gym management

Languages